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9702/32

October/November 2007

2 hours

Additional Materials: As listed in the Confidential Instructions.

DO **NOT** WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

All questions in this paper carry equal marks.

For Examiner's Use	
1	
2	
Total	

This document consists of **9** printed pages and **3** blank pages.

- 1 In this experiment, you will determine the resistance of an unknown resistor R_1 .
- (a) A wire XY has been taped to a metre rule. Connections to this wire may be made using crocodile clips.
- (i) Connect the circuit shown in Fig. 1.1. R_1 has been labelled. R_2 may be made using any series or parallel combination of the remaining resistors. These resistors may be connected to each other by twisting the resistor wires together.

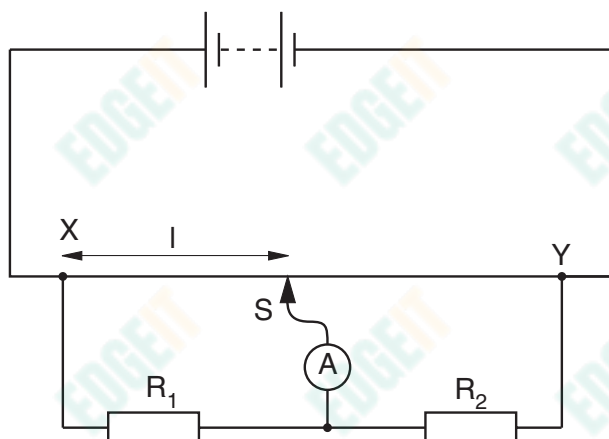


Fig. 1.1

- (ii) Position the sliding contact S on the wire and adjust the position of the contact until the reading on the meter is zero. Record the length l in centimetres from X to S.

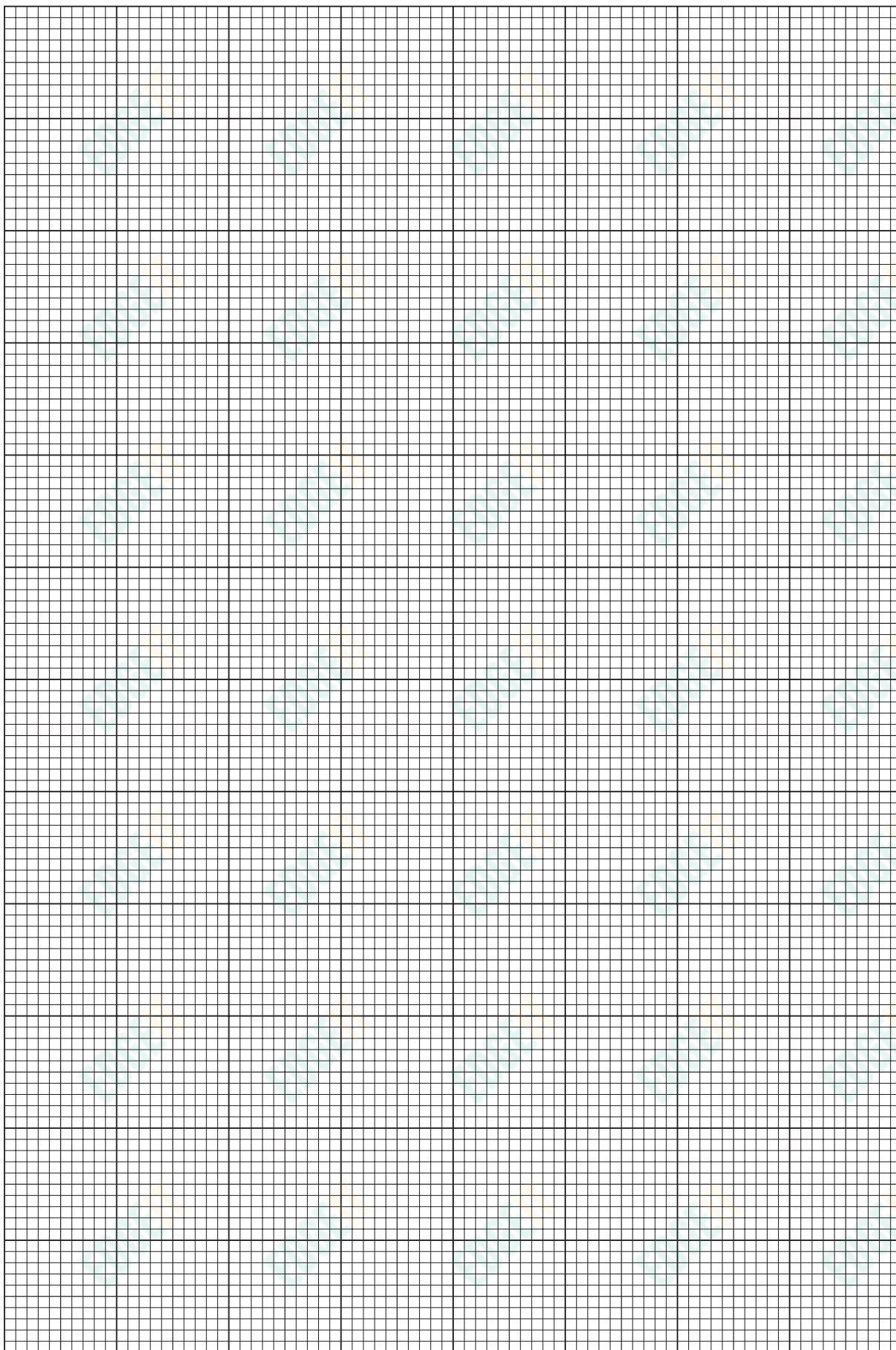
$l = \dots\dots\dots$ cm

- (b) Change the value of the resistance R_2 of the resistor R_2 and repeat (a)(ii) until you have six sets of readings for I and R_2 . Values of I should be given in centimetres. Include values of $\frac{1}{I}$ in your table of results. When you have finished making measurements, disconnect the battery.

- (c) (i) Plot a graph of $\frac{1}{I}$ (y-axis) against R_2 (x-axis).
(ii) Draw the line of best fit.
(iii) Determine the gradient and the y-intercept of the graph.

gradient =

y-intercept =



(d) The relationship between I and R_2 is

$$\frac{1}{I} = \frac{R_2}{LR_1} + k$$

where R_1 is the resistance of the resistor R_1 , $L = 100 \text{ cm}$ and k is a constant.
Using your answers from (c)(iii), determine values of R_1 and k . Include appropriate units in each case.

$R_1 = \dots\dots\dots$

$k = \dots\dots\dots$

- 2 When an object falls in air, it experiences a drag force which opposes the motion of the object. Larger objects experience greater drag forces. In this experiment, you will investigate how the terminal velocity of a paper cone falling in air depends on the diameter of the cone.

(a) Cut a sector out of a piece of filter paper as shown in Fig. 2.1.

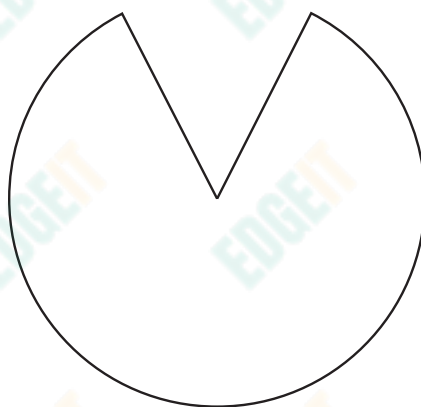


Fig. 2.1

- (b) (i) Tape the straight edges of the paper together to produce a cone, as shown in Fig. 2.2.

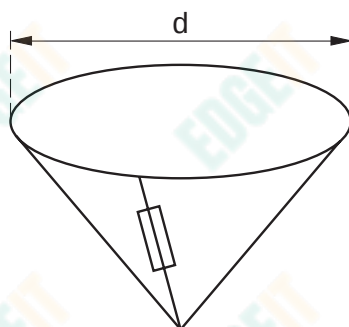


Fig. 2.2

- (ii) Measure and record the diameter d of the cone.

$d = \dots\dots\dots$ cm



- (c) (i) Mount a metre rule vertically using a stand, boss and clamp.
- (ii) Release the cone from a short distance above the top of the metre rule, as shown in Fig. 2.3.

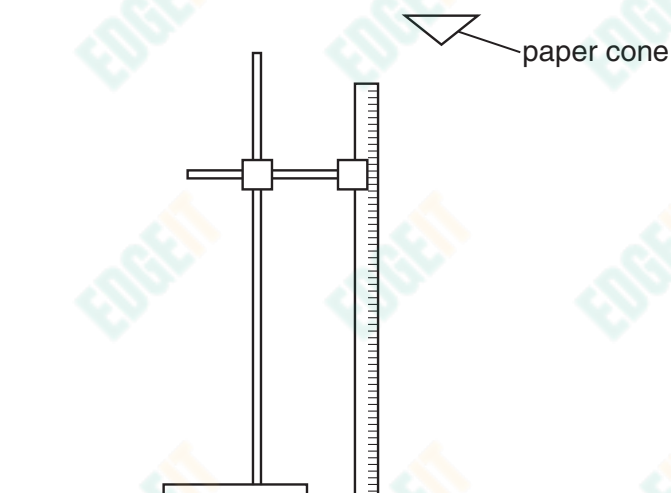


Fig. 2.3

Make and record measurements to determine the time t for the cone to fall through a distance h from the top of the metre rule.

$h = \dots\dots\dots \text{cm}$

$t = \dots\dots\dots \text{s}$

- (d) Estimate the percentage uncertainty in t , showing your working.

percentage uncertainty in $t = \dots\dots\dots$

- (e) Calculate the terminal velocity v of the cone.

$v = \dots\dots\dots \text{cm s}^{-1}$

- (f) (i) Remove the tape from the paper and cut away a larger sector as shown in Fig. 2.4.

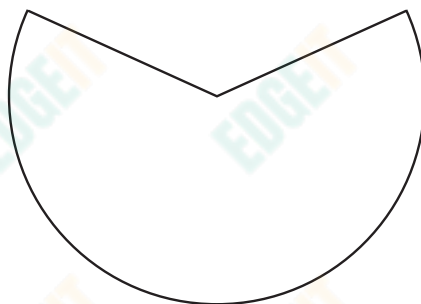


Fig. 2.4

- (ii) Repeat (b), (c)(ii) and (e), recording your results below.

$d = \dots\dots\dots \text{cm}$

$h = \dots\dots\dots \text{cm}$

$t = \dots\dots\dots \text{s}$

$v = \dots\dots\dots \text{cm s}^{-1}$

- (g) It is suggested that v is inversely proportional to d . Do the results of your experiment support this suggestion? Explain your reasoning clearly.

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(h) (i) State four sources of error or limitations of the procedure in this experiment.

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2.
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3.
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4.
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(ii) Suggest four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

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